Young's Modulus

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Aim of the Experiment

To determine the Young's Modulus (YM) of the given material through uniform and non-uniform bending by pin-microscope (PM) method.

Apparatus Required

(i) Knife Edges.

(ii) Slotted weights.

(iii) Material Bar.

(iv) Pin & Microscope

(v) X-Y translation stage.

Fig-1(a): Experimental setup for determination of YM through non-uniform bending by PM method.

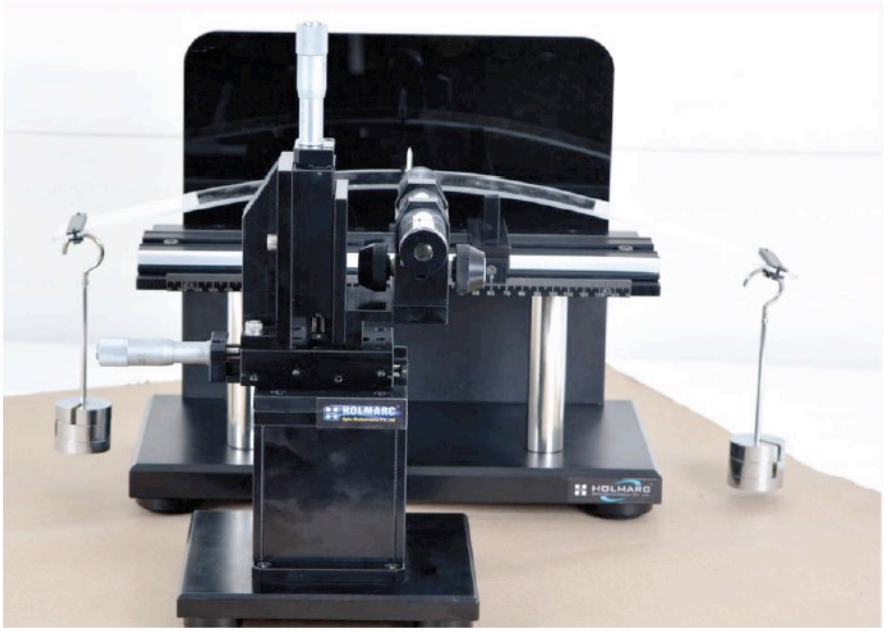

Fig-1(b): Experimental setup for determination of YM through uniform bending by PM method.

Theory

The modulus of elasticity or Young's modulus is a mechanical property of solid material that describes its stiffness. *Stiffness* is defined as the rigidity of an object to which it resists deformation in application of an external force. Every material undergoes elastic deformation by the application of external load which is a temporary deformation of the physical shape of the material and it returns to its original shape after removal of the load. This is called *elastic deformation* of materials. With the application of force F through the load, the inter-atomic distance (a) increases by δa and the material gets stretched. After removal of the force the inter-atomic force is attractive and the material regains its original shape. If we apply a force large enough so that some of the inter-atomic bond breaks and the material does not return to its original shape. This is called *plastic deformation*.

Further increase in force leads to breaking of materials. A stiff material needs more force to be deformed as compared to a soft material and an infinite force would be needed to deform a perfectly rigid material with an infinitesimal value of Young's modulus.

Let us consider a bar shaped material of length ***L*** and cross section A to which a longitudinal force ***F*** is applied. Upon deformation, the material stretches by δL. If we go on increasing the strength of this force and measure δL, the relation between F/A and $\delta L/L$ can be represented graphically as shown in *Figure-2a*. The quantity F/A is called tensile stress and $\delta L/L$ is called the tensile strain. In *Figure-2a*, the point E is the elastic limit or the yield point of the material within which the stress is proportional to strain and the material regains its original shape after removal of the external force. Beyond this point the linear relationship between stress and strain does not hold and the material becomes plastic up to the point P. Beyond the point P the material breaks. The ratio between stress to strain in the elastic region is defined as Young's modulus of the material. It is also dependent upon temperature and pressure.

$$Y = \frac{F/A}{\delta L/L}$$

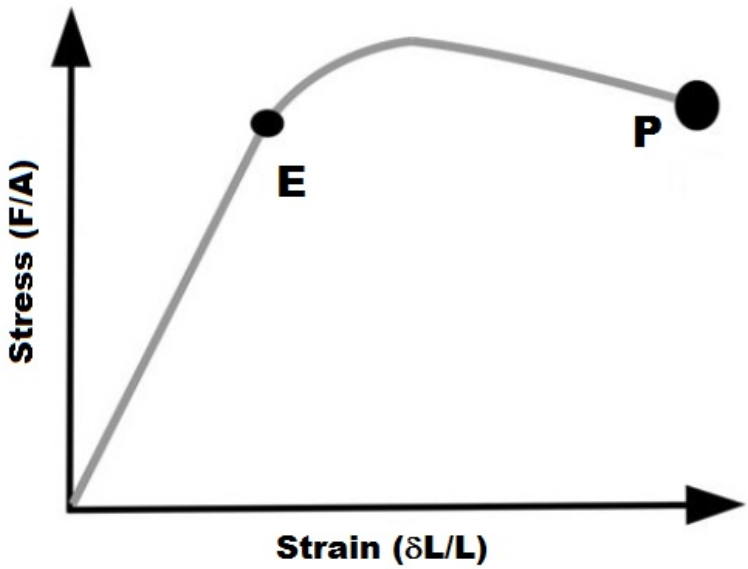

Figure-2a: Graph between stress and strain for a solid material.

Let us consider a rectangular beam of length ***L***, breadth ***b*** and thickness ***d*** supported by two knife edges and at the middle a weight is hung (*Figure-2b*). The weight at the beam gives a downward force ***W=Mg***. Since the beam is static this force is balanced by the normal reactions of magnitude ***W/2*** at the knife edges. The centre of the beam does not suffer any deflection and we can consider the entire beam as two cantilevers supported at one end and loaded at the other. This is shown in *Figure-2c*. Now let us denote the length of the cantilever is **L/2=L_1** and weight ***W/2=W_1***. Now the cantilever suffers a depression of ***z*** at the point C and needs to be calculated.

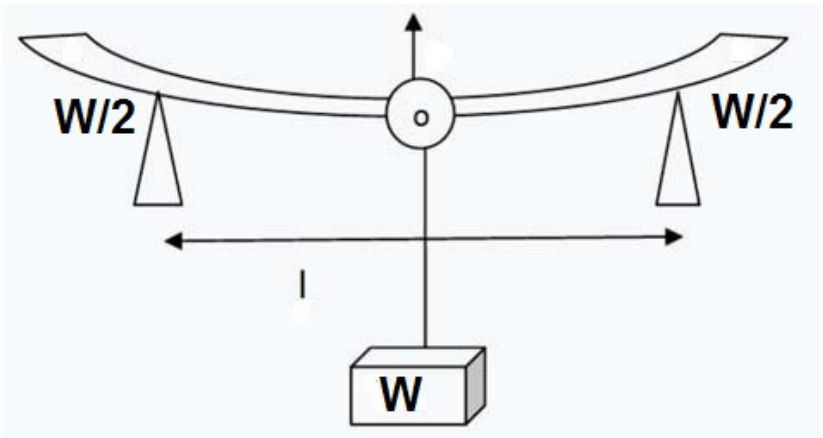

Figure-2b: Schematic diagram of the beam loaded with weight W in non-uniform bending.

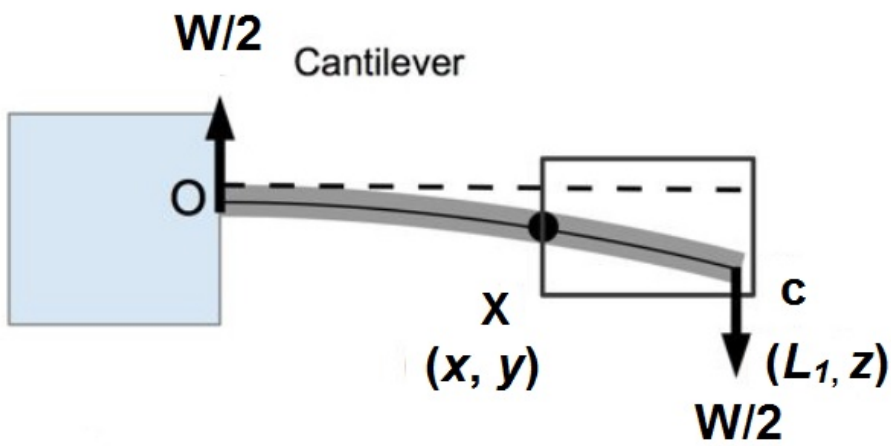

Figure-2c: Forces acting on the cantilever.

Now consider an infinitesimal section of the rectangular bar MNOP bent in the form of a circular arc of radius R. A section perpendicular to the plane of bending is shown in *Figure-2d*. As per the figure the layers of the material of the bar in the lower half will be compressed while those in upper half will be expanded. But, the middle layer remains unchanged in length is called as neutral plane and passes through the centre of gravity. In the figure XY is the neutral layer and the two ends of the layer are at an angle ϕ. Now XY=$R\phi$. As per the figure the upper layer of the neutral layer i.e. PQ is at a distance k from it which expands as the bar bends. If we draw a line parallel to MN at the point Y, it will intersect PQ layer at the point S. From geometry, the extension of this layer corresponding to the neutral layer is SQ.

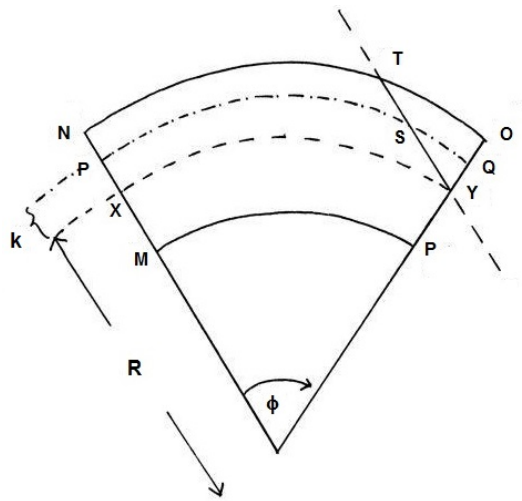

Figure-2d: An infinitesimal section of the cantilever showing the neutral layer.

$$
\begin{aligned}
SQ &= PQ - PS \\
&= PQ - XY \\
&= (R+k)\,\phi - R\phi \\
&= k\,\phi
\end{aligned}
$$

The strain developed in the extended layer will be equal to the ratio of original length XY to the extension SQ.

So strain (s) = ***k/R***

The stress produced in the material due to this expansion can be calculated using Hook's law. Consider a region of ***k*** to ***k*** + ***dk*** near to the layer PQ. Now the stress produced in that layer will be the ratio of the force $\boldsymbol{dF_r}$ produced in the material to the area ***bdk***. So we get

$$\frac{dF_r}{bdk} = Y\frac{k}{R}$$

where ***Y*** is the Young's modulus of the material. The moment of this force about the neutral layer will be

$$M = kdF_r = \frac{Yb}{R}k^2dk$$

To calculate the total moment developed about the neutral axis in the cantilever near to the point X, we need to integrate above for all possible values of ***k***, hence,

$$M = Yb/R\int_{-d/2}^{d/2} k^2\,dk = \frac{Ybd^3}{12R}$$

It is clear that the entire cantilever do not form like the part of an arc. At different parts of the beam, it will have different curvature and hence different radius of curvature. Hence, we must write

$$M(x) = \frac{Ybd^3}{12}\frac{1}{R(x)}$$

where *x* is length measured along the beam, it has a values *0* and L_1 at the point of suspension of the beam and at the loading end respectively. If we consider the depression at different point in the beam to be y, then *y=0* at the point O and *y=d* at the point C. We have *dy/dx = 0* at C. So we can write the radius of curvature of the beam R(x) as

$$\frac{1}{R} = \frac{d^2y}{dx^2}$$

Putting this value in the above eqn. we got,

$$M(x) = \frac{Ybd^3}{12}\frac{d^2y}{dx^2}$$

M(x) gives the internal bending moment of the beam at the position x and this should be same as the external bending moment provided by the weight and the normal reaction. Clearly from the Figure-2c the bending moment at A because of the weight W_1 at C would be (neglecting the weight of the rod) $W_1 (L_1 - x)$, hence,

$$W_1(L_1 - x) = \frac{Ybd^3}{12}\frac{d^2y}{dx^2}$$

We have already stated the boundary conditions for this differential equation. Solving this we may get the value of z at $x = L$, $y = z$,

$$z = \frac{4W_1L_1^3}{Ybd^3}$$

But, in the above case we have only chosen the half of the beam. Hence, the depression z produced in the entire beam loaded at the middle, in terms of W and L, will be

$$z = \frac{WL^3}{4Ybd^3}$$

In the experiment, we can measure the dimensions of the beam (i.e L, b, d) and the depression for different applied weight W. Hence we can invert the above equation to estimate the Young's modulus from the experiment as

$$Y = \frac{WL^3}{4zbd^3} = \frac{MgL^3}{4zbd^3}$$

This is the working formula for this experiment for non-uniform bending.

For uniform bending (*Figure-2e*) equal weights Mg is suspended at both the ends of the material bar at an equal distance p from each knife edges. So elevation at the midpoint is given by z where

$$z = \frac{3WpL^2}{2Ybd^3}$$

So the Young's modulus can be written as,

$$Y = \frac{3WpL^2}{2zbd^3} = \frac{3MgpL^2}{2zbd^3}$$

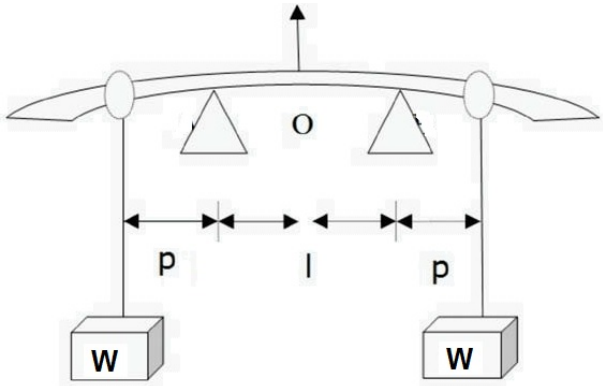

Figure-2e: Schematic diagram of the beam loaded with weight W in uniform bending.

Procedure

Non-uniform bending by PM method

(i) Place the bar is symmetrically on two knife edges. Fix the distance between the knife edges. The weight hanger is suspended from the centre.

(ii) Fix the pin vertically at the midpoint of the bar with its pointed end upwards. Load and unload the bar a number of times so as to bring it into the elastic mood.

(iii) Arrange the microscope in front of the pin and focus the tip of the pin. Coincide the intersecting point of horizontal and vertical cross wire of the microscope with the tip of the pin with dead load or zero load.

(iv) Record the readings on the vertical scale and vernier of the microscope. Now load 'M' kg to the weight hanger.

(v) Adjust the microscope again so that the intersecting point of its horizontal cross wire coincides with the tip of the pin. Record the microscope reading again.

(vi) Determine the mean depression 'z' of the midpoint of the bar for a given mass from these readings. Measure the breadth 'b' and thickness 'd' of the bar using vernier callipers and screw gauge respectively. Calculate the Young's modulus of the given material of the bar using the given equation.

Uniform bending by PM method

(i) Place the bar is symmetrically on two knife edges. Suspend the two weight hangers at the extreme points and at equal distance from the knife edges.

(ii) Load and then unload the bar with weights a number of times so as to bring it into the elastic mood. Fixapin vertically at the centre of the bar. Place the microscope in front of the pin and focus the crosswire at the tip of the pin.

(iii) Coincide the intersecting point of the horizontal and vertical cross wire of the microscope with the tip of the pin with dead load or zero load alone. Record the readings on the vertical scale and vernier of the microscope.

(iv) Load 'M' kg to the weight hanger and adjust the microscope so that its horizontal cross wire coincides with the tip of the pin.

(v) Measure the length of the bar 'l' between the knife edges, breadth 'b' and the distance of the load 'p' from the knife edge.

(vi) Determine the mean elevation 'z' of the midpoint of the bar for a given mass from these readings. Calculate the Young's modulus of the given material of the bar using the given equation.

Observations

Vernier constant of micrometer=0.01 mm

Distance of knife edge (L)=20 cm

Table-1: Non-uniform bending

No of Obs.	Distance of the knife edges (L)	Load (M) in Kg	Telescope reading			Depression for load 4m (z) in cm	Mean Z in cm	L^3/z in cm^3	Mean L^3/z cm^3
			Loading	Unloading	Mean				
1	Aluminium	0	10.40	10.39	10.395	0.0235	0.025625	340425.5319	313083.5277
		0.1	10.35	10.33	10.340	0.0255		313725.4902	
		0.2	10.29	10.26	10.275	0.0265		301886.7925	
		0.3	10.23	10.21	10.220	0.0270		296296.2963	
		0.4	10.16	10.16	10.160				
		0.5	10.10	10.07	10.085				
		0.6	10.01	10.01	10.010				
		0.7	9.95	9.95	9.950				
2	Brass	0	7.68	7.68	7.680	0.0205	0.018875	390243.9024	425217.9559
		0.1	7.60	7.62	7.610	0.0190		421052.6316	
		0.2	7.56	7.56	7.560	0.0185		432432.4324	
		0.3	7.52	7.53	7.525	0.0175		457142.8571	
		0.4	7.47	7.48	7.475				
		0.5	7.41	7.43	7.420				
		0.6	7.37	7.38	7.375				
		0.7	7.35	7.35	7.350				

Table-2:uniform bending

No of Obs.	Distance of the knife edges (L)	Load (M) in Kg	Telescope reading			Depression for load 4m (z) in cm	Mean Z in cm	L3/z in cm3	Mean L3/z cm3
			Loading	Unloading	Mean				
1	Aluminium	0	9.92	9.87	9.895	0.106	0.1025	3773.585	3904.181
		0.1	10.19	10.18	10.185	0.102		3921.569	
		0.2	10.46	10.46	10.460	0.100		4000.000	
		0.3	10.69	10.71	10.700	0.102		3921.569	
		0.4	10.96	10.95	10.955				
		0.5	11.21	11.20	11.205				
		0.6	11.45	11.46	11.455				
		0.7	11.72	11.72	11.720				
2	Brass	0	7.65	7.64	7.645	0.0605	0.06225	6611.570	6428.296
		0.1	7.80	7.78	7.790	0.0625		6400.000	
		0.2	7.95	7.94	7.945	0.0620		6451.613	
		0.3	8.11	8.09	8.100	0.0640		6250.000	
		0.4	8.25	8.25	8.250				
		0.5	8.42	8.41	8.415				
		0.6	8.56	8.57	8.565				
		0.7	8.74	8.74	8.740				

Calculation

	Aluminium	**Brass**
Thickness of the material of the bar "d"	= 2.39 mm	2.33 mm
Breadth of the material of the bar "b"	= 5 cm	5 cm

Load applied for depression "z" (=4M) = 0.4 kg

Distance between knife edges and suspended mass, p(for uniform bending)= 15 cm

For material Aluminium Young's Modulus, Y= $\frac{\mathbf{Mg}L^3}{\mathbf{4zb}d^3}$ **(for non uniform method)**

$$=\frac{0.4*9.8*313083.5277*10^{-4}}{4*5*10^{-2}*(2.39)^3*10^{-9}}\ \mathrm{N/m^2}$$

$$=44.94*10^9\ \mathrm{N/m^2}\ \text{or}\ 44.94\ \mathrm{GPa}$$

Y= $\frac{\mathbf{3Mgp}L^3}{\mathbf{2zb}d^3}$ **(for uniform method)**

$$=\frac{3*0.4*9.8*0.15*3904.181*10^{-2}}{2*5*10^{-2}*(2.39)^3*10^{-9}}\ \mathrm{N/m^2}$$

$$=50.44*10^9\ \mathrm{N/m^2}\ \text{or}\ 50.44\ \mathrm{GPa}$$

For material Brass Young's Modulus, Y= $\frac{\mathbf{Mg}L^3}{\mathbf{4zb}d^3}$ **(for non uniform method)**

$$=\frac{0.4*9.8*425217.9559*10^{-4}}{4*5*10^{-2}*(2.33)^3*10^{-9}}\ \mathrm{N/m^2}$$

$$=65.8810^9\ \mathrm{N/m^2}\ \text{or}\ 65.88\ \mathrm{GPa}$$

Y= $\frac{\mathbf{3Mgp}L^3}{\mathbf{2zb}d^3}$ **(for uniform method)**

$$=\frac{3*0.4*9.8*6428.296*0.15*10^{-2}}{2*5*10^{-2}*(2.33)^3*10^{-9}}\ \mathrm{N/m^2}$$

$$=89.64*10^9\ \mathrm{N/m^2}\ \text{or}\ 89.64\ \mathrm{GPa}$$

Precautions

(i) Put the material bar carefully on knife edges.

(ii) Give sustainable weights to the material bar.

Considerations:

(i)

- Stress is a physical quantity that expresses the material exert on each other, while strain is the measure of the deformation of the material. Stress can be defined as the traction force acting on a small cross section of the body per unit area.

- Strain is usually defined as the change in some dimension of the body divided by its original dimensions

(ii)

- Elasticity is the ability of a body to resist a distorting influence and to return to its original size and shape when that influence or force is removed. Solid objects will deform when adequate loads are applied to them; if the material is elastic, the object will return to its initial shape and size after removal.
- Plasticity, also known as plastic deformation, is the ability of a solid material to undergo response to applied forces.

(iii)

- The yield point is the point on a stress-strain curve that indicates the limit of elastic behaviour and the beginning of plastic behaviour. Below the yield point, a material will deform elastically and will return to its original shape when the applied stress is removed.
- Hooke's law can be simply stated as a linear proportionality between stress and strain and the constant of proportionality is a constant for a given material and is called Youngs' modulus of elasticity.

Conclusions

(i) **Interesting results you found in the experiment:** Young's modulus is meaningful only in the range in which the stress is proportional to the strain, and the material returns to its original dimensions when the external force is removed. As stresses increase, the material may either flow, undergoing permanent deformation, or finally break.

(ii) **Lacunas /difficulties that you encountered:** This experiment serves as an crude method to estimate young's modulus for a material beam. The values obtained are obtained accurate to the order and to within 10GPa of the actual quoted values. The large error can be attributed to the fact that the change in dimensions and deflections are extremely small and measuring them accurately is hard even by using micrometer screws. Another source of error is that the behaviour of the beam is also affected by its previous history of deformation and residual stresses. In the modern research setting many complicated and expensive machines already exist in the field that allows us to measure such tiny deformations and accurately measure other material properties.

(iii) Practical applications/implications of the experiment. Studying of the mechanical properties of materials is important because it helps us understand how materials behave, and allows us to develop new products and improve existing ones. Knowledge of different material properties is of utmost interest and importance to various fields like civil, architecture, materials engineering, designing and research. Behaviour of materials under stress or deformation finds various applications in other areas as well. A good example of this would be that piezoelectricity and other exotic materials like nitinol are quickly finding new uses in modern technology.